Experiment Targets

Planning guide for *After the Decision*

These are **illustrative research goals, not forecasts or observations**. Recorded comparators remain unchanged. Blue T rows show an arithmetically consistent target scenario, normalized to existing denominators. Statistical success and generalization still require real evaluation.

Group	Body output	Research question	Goal checkpoint (T)
E1	Table 3 / Fig. 3	Natural gain and delayed recovery	$+1.02$ / $+1.23$ pp*
E2	Table 4	Correct source attribution	Local beats permutation
E3	Table 5	Learnable cold-start headroom	81.86 / 81.72%
E4	Table 6	Value of the update alone	$+1.39$ / $+1.20$ pp*
E5	Table 7	Cost at one call per input	Selection $\leq 30/35\,\mu$s
E6	Table 8	Refusal at matched coverage	95.78%; $U = -0.0845$

* Natural / controlled gain versus current Relay in E1, current corrected Query in E4. These E-labels match the presentation draft; the active standard's E6 means untouched confirmation.

1. Primary selection and delay

Four providers; 157 molecules / 124 scaffold clusters; $D \in \{0, 1, 4\}$. Each method has 4,710 calls across 30 schedules, *not* 4,710 independent molecules. Controlled drift is an evaluator-side label flip after tick 78.

Table 1: E1 / body Table 3. Recorded anchors and planning target (T).

	Natural menu		Controlled drift	
Method / status	Correct	Success (%)	Correct	Success (%)
Random (M)	3,945	83.76	—	—
Best Fixed (M)	4,152	88.15	—	—
Frozen (M)	4,230	89.81	3,456	73.38
MIRA-Relative (M)	4,230	89.81	—	—
Online SGD (M)	4,202	89.21	3,958	84.03
Greedy LinUCB (M)	4,226	89.72	3,822	81.15
BayesianRouter (M)	4,217	89.53	3,781	80.28
Current MIRA-Relay (M)	4,230	89.81	4,124	87.56
Retrospective Best Fixed† (M)	4,260	90.45	—	—
Improved MIRA-Relay (T)	4,278	90.83	4,182	88.79

† Retrospective Best Fixed chooses one fixed provider with evaluator truth; it is not an upper bound on contextual selection. The natural per-molecule oracle is 150/157 = 95.54%, or 4,500 recorded decisions. T adds 48 correct natural decisions: 17.8% of Frozen's 270 recoverable decisions.

Table 2: E1 / body Figure 3 data. Gains over Frozen, in pp; T is illustrative.

	Natural menu		Controlled drift	
Delay	Current (M)	Target (T)	Current (M)	Target (T)
0	-0.13	$+1.15$	$+13.63$	$+15.67$
1	-0.13	$+1.02$	$+14.52$	$+15.41$
4	$+0.25$	$+0.89$	$+14.39$	$+15.16$

Each delay has 1,570 calls. T correct counts: natural 1,428/1,426/1,424; controlled 1,398/1,394/1,390. Their sums reproduce Table 1. A compatible aggregate rescue/harm scenario is 110/62 (natural) and 818/92 (controlled), giving net 48 and 726 versus Frozen. This is not a predicted event trace.

Decision test. Seek a practically meaningful natural gain and retain controlled recovery; require prospectively specified paired scaffold uncertainty against the strongest legal matched control. Target totals alone cannot demonstrate significance. Delay points are goals, not a fitted law.

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2. Mechanism and learnable headroom

E2 preserves initial scores, source residual computation, budget and release clock; only the credit destination changes. Its scaffold-equal estimand differs from E1's pooled point estimates.

Table 3: E2 / body Table 4. Source-credit contrasts (pp); goals are dependent.

	Natural		Controlled	
Contrast	Current (M)	Target (T)	Current (M)	Target (T)
Local − Frozen	+0.10	+1.10	+14.40	+15.50
Permutation − Frozen	+0.41	+0.10	−1.27	−1.00
Local − permutation	−0.30	+1.00	+15.67	+16.50

For T: $1.10 - 0.10 = 1.00$ and $15.50 - (-1.00) = 16.50$. The three contrasts are not independent successes. Measured contrasts use unrounded values before subtraction. Actual paired intervals must support the attribution claim; do not derive intervals from these target means.

Table 4: E3 / body Table 5. Static cold-start targets on 722 development molecules.

Selector / status	Correct / 722	Accuracy (%)
Best Fixed (M)	572	79.22
Raw Frozen (M)	573	79.36
Corrected Frozen (M)	579	80.19
BCE + CLS (M)	567 / 570	78.53 / 78.95
BT + CLS (M)	569 / 565	78.81 / 78.25
Improved prior (T)	591 / 590	81.86 / 81.72
Per-molecule oracle† (M)	618	85.60

Paired seeds 0/1 use the same molecules; '/' separates seed results. The oracle is evaluator-only. Of 722 molecules, 471 have all providers correct, 147 are mixed and 104 are all wrong. T captures 12/11 of the 39 remaining selection opportunities (30.8/28.2%). It needs a better ranking representation or training rule; current learned priors have not demonstrated this effect.

Decision test. Both seeds should exceed the corrected Frozen by at least 1 pp, without negative leave-one-fold-out contrasts. Static ranking success alone does not prove online adaptation.

3. Isolate the update

Table 5: E4 / body Table 6. Score × update diagnostic; 2,166 calls per cell.

	Natural		Controlled	
Score / update / status	Correct	Accuracy (%)	Correct	Accuracy (%)
Raw / Frozen (M)	1,719	79.36	1,497	69.11
Raw / Marginal (M)	1,715	79.18	1,674	77.29
Raw / Query (M)	1,700	78.49	1,645	75.95
Corrected / Frozen (M)	1,737	80.19	1,371	63.30
Corrected / Marginal (M)	1,729	79.82	1,696	78.30
Corrected / Query (M)	1,740	80.33	1,704	78.67
Corrected / improved update (T)	1,770	81.72	1,730	79.87

The T update keeps the *same corrected prior as measured E4*; it does not use the E3 target prior. T gains over Query are +30/+26 correct calls (+1.39/+1.20 pp); gains over Marginal are +41/+34 (+1.89/+1.57 pp). Natural T stays below the fixed-menu oracle cap of $618 \times 3 = 1,854$. Each policy generates its own selected-feedback history.

Decision test. Require the 1 pp development resource line and nonnegative leave-one-fold-out contrasts within each fixed score scale. Cross-scale differences also change intervention exposure; they cannot isolate the update's benefit.

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4. Retain efficiency while adding useful capacity

E5 caps preserve room for an improved mechanism. They are engineering budgets, not predictions that every latency will decrease.

Table 6: E5 / body Table 7. Mean of stream-level p95s (μs); T rows are caps.

	Natural		Controlled	
Method / status	Selection	Update	Selection	Update
Frozen (M)	15.89	1.31	17.88	1.50
Online SGD (M)	14.08	19.37	14.29	20.02
Greedy LinUCB (M)	128.63	90.90	128.36	92.32
BayesianRouter (M)	133.65	88.96	137.34	92.55
Current MIRA-Relay (M)	27.21	2.81	32.47	3.46
Improved module (T; caps)	≤ 30	≤ 5	≤ 35	≤ 5

Same macOS arm64 measurement scope; mean of 30 stream p95s, not a pooled p95. Excludes provider forward, encoding, I/O and receipt bookkeeping. One call per input remains fixed. If the 5 μs cap is met, updating would be at least $19.37/5 = 3.87\times$ faster than the current natural SGD anchor *for this kernel metric only*. Live learner memory and full-stack latency must be measured; no numerical forecast is justified yet.

Decision test. Meet the efficacy target and these kernel caps jointly. Report encoding, pending/history management, memory and recovery separately before making deployment-speed claims. Extra capacity must not silently acquire additional labels or tool calls.

5. Reduce accepted errors without reducing coverage

E6 is a secondary analysis of the inspected cohort: five fit-only policies, each applied to 157 molecules ($N = C = 785$ records/calls per policy). It is not new untouched confirmation.

Table 7: E6 / body Table 8. Matched-count refusal target; T is not an observed policy.

Policy / status	Accepted	Wrong	Refused	Coverage	Acc.*	Utility
Always accept (M)	785	93	0	100.00	88.15	-0.1285
Confidence refuse (M)	663	36	122	84.46	94.57	-0.0947
Always refuse (M)	0	0	785	0.00	—	-0.2600
Same-coverage random† (M)	663	75	122	84.46	88.69	-0.1444
Improved ranking (T)	663	28	122	84.46	95.78	-0.0845

* Accepted accuracy; coverage and accuracy in %. † Evaluator-only conditional random null. $U = -(W + 0.25R + 0.01C)/N$, where W is accepted errors and R refusals. Always-refuse accuracy is undefined, not zero. T implies 635 correct accepted outputs and $U = -(28 + 0.25 \times 122 + 0.01 \times 785)/785 = -0.0845$.

Decision test. The target requires replacing eight accepted wrong predictions with eight correct ones, at 663 accepted outputs; the existing unchanged threshold cannot do this. Freeze a revised fit-only ranking and report risk–coverage and cost sensitivity. Exact matched coverage is an analytical control, not guaranteed prospective threshold behavior. Utility gain: +0.0102 versus current confidence refusal; +0.0440 versus always accept.

6. What would turn these goals into evidence?

1. Close the natural-selection and attribution gaps first. Isolate E3 prior and E4 update effects, then combine them under a matched budget to establish one coherent method.
2. Freeze the candidate, comparators and statistical plan before new evaluation. The inspected 157 molecules are not untouched confirmation; transfer target rates, not old counts.
3. Retain negative results and independent recovery. Endpoint, lifecycle and agent extensions need separate evidence. The full scientific gate remains **HOLD**.

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Sources (paths/hashes in target_spec.json): E1/E5 fresh natural/stress; E2 mechanism v2; E3 headroom/learned-prior; E4 calibration v1; E6 corrected post-call audit v3. Real-paper tables remain unchanged.

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